Fiscal Year:	FY 2014	**Task Last Updated:**	FY 08/11/2014
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Project Title:	Novel Double-Hit Mouse Model to Investigate Oxidative Damage from Radiation/ Hyperoxia Related to Space Exploration - Evaluation of Dietary Flaxseed as Countermeasure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	06/01/2012	**End Date:**	05/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2014 (from 5/31/2013) per A. Chu/ARC (Ed., 3/15/2013)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX12AK19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight missions may require crewmembers to conduct extravehicular activities (EVA) for repair, maintenance, or scientific purposes. EVAs should total no more than 24 hours weekly per crewmember. During each EVA, crewmembers are subjected to a number of challenges that may induce oxidative tissue damage and thus present a threat to their health and therefore pose a limitation to the success of the mission. One such challenge is the exposure to 100% oxygen as well as to low levels of total body cosmic/galactic radiation that crewmembers will be exposed to during an EVA. Currently, no studies have addressed this “double hit” hyperoxia/radiation effect so we developed and characterized a novel, mouse model of repeated total body radiation and hyperoxia exposure. Preliminary studies identified acute lung damage associated with such treatment so the goal of the proposed study is to further characterize the acute damage and most importantly, to provide information on the long-term effects of such an exposure (Aim 1). Additionally, initial findings suggested that dietary administration of whole grain dietary flaxseed (FS) ameliorated adverse acute effects in the same model. Our group was the first to show in murine models of radiation and hyperoxic lung injury the protective properties of FS. Importantly, lignan phenolics in FS such as secoisolariciresinol diglucoside (SDG), are known antioxidant, anti-inflammatory, and anticarcinogenic agents. We hypothesize that FS and FS lignans will be effective countermeasures of this double-hit toxicity to lung tissues (Aim 2). We plan to further explore the usefulness of FS and SDG in both acute and chronic oxidative lung damage in this model. FS and specifically SDG are safe, readily available and thus, attractive candidates to evaluate as oxidative damage countermeasures in the proposed double hit radiation/hyperoxia model. Data from this work are relevant to EVA-related pathophysiology associated with space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We aim to identify novel risks to lung tissues as a result of exposure to hyperoxia and radiation. This is relevant to both space travel and clinical Earth scenarios. We hope that our proposed countermeasure will mitigate risk.
Task Progress:	BACKGROUND Spaceflight missions may require crewmembers to conduct extravehicular activities (EVA) for repair, maintenance, or scientific purposes. Pre-breathe protocols in preparation for an EVA entail 100% hyperoxia exposure that may last for a few hours (5-8 hours) and be repeated 2-3 times weekly. Each EVA is associated with additional challenges such as low levels of total body cosmic/galactic radiation exposure that may present a threat to crewmember health and therefore, pose a threat to the success of the mission. We have previously characterized the lung tissue changes in a murine model of repeated radiation and hyperoxia exposure (Aim 1). Flaxseed (FS), and more specifically flaxseed lignan complex enriched in the lignan secoisolariciresinol (FLC), has lung protective properties in mouse models of acute and chronic lung injury. To test the usefulness of FS as a dietary supplement and countermeasure for such missions (Aim 2), we evaluated FS in an acute and chronic murine model of double-hit hyperoxia and radiation exposure. METHODS Acute Model: In order to determine acute effects of repeated radiation and hyperoxia exposure, we evaluated lung tissue changes 2 weeks post-initiation of exposure challenges. Mouse cohorts (n=5/group) were pre-fed diets containing either 0% FS or 10% FS for 3 weeks and exposed to: a) normoxia (Untreated); b) >95% O2 (O2); c) 0.25Gy single fraction gamma radiation (IR); or d) a combination of O2 and IR (O2+IR). The stress challenge (O2, IR, or a combination of both) was given 3 times per week for 2 consecutive weeks and consisted of 8-hour hyperoxia treatments spanned by normoxic intervals. Cohorts that received IR were exposed to gamma radiation shortly prior to entering the hyperoxic chambers. Lungs were evaluated for tissue damage, oxidative and nitrosative stress, inflammation, injury, and fibrosis after 2 weeks of exposure. Chronic Model: In order to determine late effects of repeated radiation and hyperoxia exposure, we evaluated lung tissue changes 16 weeks post-initiation of exposure challenges. Additionally, along with testing whole grain FS, we evaluated the usefulness of the flaxseed lignan component (FLC), enriched in lignans such as secoisolariciresinol diglucoside (SDG), as countermeasures of this double-hit toxicity to lung tissues. Specifically, mouse cohorts (n=5/group) were fed 0% FS, 10% FS, or 10% FLC and exposed to all challenge conditions, as previously described, for 4 weeks, i.e., 12 rounds of exposure. We evaluated chronic lung changes, lung inflammation, cytokine levels and lung oxidative and nitrosative stress, as well as changes in mouse bodyweight and blood PaO2. RESULTS Acute Model: At 2 weeks post challenge, control diet-fed mice (0% FS) developed significant lung injury and inflammation across all challenges, as evidenced by bronchoalveolar lavage fluid (BALF) neutrophils ($p<0.003$) and protein levels. Additionally, they showed increased oxidative tissue damage, as determined by levels of malondialdehyde (MDA), a biomarker of lipid peroxidation and oxidative stress ($p<0.008$). The adverse effects due to challenge conditions were ameliorated in 10% FS-fed mice. Mice fed 10% FS had significantly reduced levels of BALF MDA and nitrite, a stable measure of nitric oxide and a biomarker of nitrosative stress. Lung hydroxyproline levels, a measure of lung fibrosis, were significantly elevated in mice fed 0% FS ($p<0.01$) and exposed to hyperoxia/radiation or the combination treatment, but not in FS-fed mice. To determine whether FS altered levels of a pro-inflammatory, pro-fibrogenic cytokine (TGFB1) we looked at gene expression levels in lung. We determined a significant,1.61-, 1.55-, and 2.55-fold induction of TGFB1 gene expression level over control in mice fed 0% FS and exposed to O2, IR, and O2+IR, respectively. Flaxseed abrogated the challenge-induced increase in TGFB1 in all exposure conditions. (Manuscript in prep) Chronic Model: At 16 weeks post-challenge we determined blood PaO2 and lung hydroxyproline levels, at the time of sacrifice of each cohort, and observed no significant change from untreated control mice. Notably, however, mice fed 10% FS or 10% FLC improved lung inflammation, as determined by BALF WBCs. Most of the 20 cytokines analyzed in the BALF were low (below the lower limit of detection of the assay); however, IL-10, a cytokine with known anti-inflammatory properties, was significantly elevated in BALF from FS and FLC fed mice exposed to combination treatment (O2+IR). Importantly, at 16 weeks post-challenge, persistent elevations in oxidative and nitrosative stress, as determined by BALF levels of MDA and nitrite, were significantly abrogated in mice fed 10% FS or 10% FLC. (Manuscript in prep). CONCLUSION

	In conclusion, dietary FS was protective in an acute and chronic murine model of low-level radiation and hyperoxia exposure. Notably, the elevation in oxidative and nitrosative lung tissue damage brought on by hyperoxia and total body radiation exposure was blunted by FS. More importantly, the lignan component of flaxseed, enriched in secoisolariciresinol diglucoside, was comparably effective to whole grain FS in ameliorating the late negative effects of hyperoxia and radiation exposure.
Bibliography Type:	Description: (Last Updated: 10/31/2019)
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